

ENTERPRISE LOYALTY ENGINE SPECIFICATION PROPOSAL

A Multi-Brand Referral, Tier-Progression, & Gamification Ecosystem

CLIENT: BRADFORD INTERNATIONAL ALLIANCE (SHARJAH OFFICE)
DOCUMENT VERSION: 3.1 (UX & ANIMATIONS ADDED)

TABLE OF CONTENTS

1. Executive Summary & Sharjah BIA Business Context	Page 2
2. Points Accrual Basis & Event Triggers	Page 3
3. Relational Database Schema & Architecture Blueprint	Page 4
4. API Endpoint Specifications & Ledger Webhooks	Page 5
5. Shukran-Style Tier Management & Multipliers	Page 6
6. Points Expiration Algorithms & Expiry Mathematics	Page 7
7. User Experience (UX), Tech Stack & Animations (GSAP)	Page 8
8. WhatsApp Business API & LMS Integrations	Page 9
9. Security Protocols, Anti-Fraud Middleware, & Compliance	Page 10
10. Implementation Timelines (Intern + AI) & QA Test Suites	Page 11

Confidential Document. For internal BIA planning and review only. Unauthorized distribution is prohibited.

1. EXECUTIVE SUMMARY & SHARJAH BIA BUSINESS CONTEXT

Bradford International Alliance (BIA), operating in Al Nahda, Sharjah, represents a premier academic consultancy. It facilitates undergraduate, postgraduate (MBA, DBA), and Doctorate pathways for corporate professionals in the UAE. Executive academic courses carry tuition fees ranging from AED 10,000 to over AED 50,000. Due to these high transaction values and the relationship-driven nature of academic registration, word-of-mouth represents BIA's most valuable marketing channel.

A standard student referral program is simple. However, to create a system that drives high retention and long-term brand loyalty (similar to Landmark Group's Shukran program), BIA needs an integrated Multi-Brand Loyalty Engine. This engine connects BIA's primary education offerings with its subsidiaries: BIA Academics, Bradford Learning Global (professional certifications like ACCA and CMA), and BIA EduAbroad.

This specification proposal details a modular, highly scalable system designed for BIA's specific tech capabilities. It outlines a strategy where an intern, guided by Antigravity AI, can deploy a transactional custom API in 12 days. Alternatively, it presents a 5-day headless SaaS option to quickly validate client acquisition rates.

Core Program Values (As per PDF Proposal):

- Point Equivalency: 100 points = AED 25 credit (1 point = AED 0.25).
 - First-Referral Bonus: Credits 1,000 points (AED 250 value) to the referrer.
 - Subsequent Referrals: Credits 250 points (AED 62.50 value) to the referrer.
 - New Student Welcome: Credits 200 points (AED 50 value) to the referee.
 - Program Bonus: Additional 100 points (AED 25 value) if the referee joins an MBA, DBA, or Doctorate.
-

2. POINTS ACCRUAL BASIS & EVENT TRIGGERS

To move beyond a simple referral code and replicate a true enterprise loyalty program like Shukran, students must be rewarded across multiple activities. The engine calculates and allocates points based on four primary pillars:

A. Referral & Enrollment Milestones

Points are awarded when a student refers a friend who registers. Crucially, points are only credited after the referred referee completes enrollment and verifies their first installment payment. This protects BIA from shell signups.

B. Academic Milestones & LMS Activity (SkillShare)

Students earn rewards for active academic engagement. Webhooks from the BIA LMS (e.g. SkillShare) automatically trigger credits when a student finishes an extra-curricular course (100–200 base points) or passes a certification module.

C. Tuition Fees & Spend-Based Points

To reward consistent payment, BIA integrates its ERP/billing system with the loyalty subdomain. Students earn points on a spend-ratio (e.g., 1 point per 50 AED paid towards tuition). This functions as a cash-back reward program to help fund future semesters.

D. Seminar & Networking Event Attendance

Building community is vital for professional networking in Sharjah. Students who attend BIA-hosted physical seminars, guest lectures, or alumni networking galas can scan a portal check-in QR code to instantly claim 50–150 engagement points.

3. RELATIONAL DATABASE SCHEMA & ARCHITECTURE BLUEPRINT

Because loyalty points are a form of financial ledger credit, the database must be relational (SQL) to support transactions (ACID compliance). The schema below specifies the tables, fields, and indexing strategies optimized for MySQL:

Table 1: users (Tracks balances & tiers)

- user_id (INT, PK, Auto Increment) - Unique student identifier.
- name / email / referral_code (VARCHAR) - Unique fields with indexes.
- current_tier (ENUM: Bronze, Silver, Gold, Platinum) - Defaults to Bronze.
- points_balance / total_earned (INT) - Tracks current wallet balance.

Table 2: referrals (Tracks enrollment workflows)

- referral_id (INT, PK) - Unique referral tracking ID.
- referrer_id (INT, FK) - Link back to users table.
- referee_email / program_name (VARCHAR) - Tracking attributes.
- status (ENUM: Lead, Enrolled, FirstPaymentCompleted, Cancelled).
- payment_date (TIMESTAMP, Nullable) - Triggers the points release.

Table 3: points_ledger (Double-entry transaction audit log)

- ledger_id (INT, PK) - Ledger tracking index.
- user_id (INT, FK) - Link to users.
- points_change (INT) - Signed integer (+250, -500).
- event_type (ENUM: Referral, ProgramBonus, Welcome, SkillShare, Expiry, Redemption).
- expires_at (TIMESTAMP) - Automatically set to Transaction Date + 4 Years.
- points_remaining (INT) - Tracks unredeemed portion of this specific deposit.

Indexing Strategy for Performance:

- INDEX(users.referral_code): Speed up referral link lookups during signup.
- INDEX(points_ledger.expires_at, points_ledger.points_remaining): Optimizes nightly cron checks.

4. API ENDPOINT SPECIFICATIONS & LEDGER WEBHOOKS

The BIA Loyalty subdomain will expose RESTful API endpoints for backend triggers (LMS, payment portal) and student frontend wallets. All requests must utilize Bearer token authorization:

1. POST /api/v1/loyalty/referral-lead

- Purpose: Records a new referral submission when a student enters a referral code.
- Request Payload:

```
{ "referrer_code": "SARAH-9042", "referee_email": "john@email.com", "program": "MBA" }
```
- Response (201 Created):

```
{ "success": true, "message": "Referral registered successfully", "referral_id": 404 }
```

2. POST /api/v1/loyalty/payment-verified

- Purpose: Triggers points release once the finance portal verifies the first payment.
- Request Payload:

```
{ "referral_id": 404, "invoice_number": "INV-2026-092", "amount_paid": 5000 }
```
- Response (200 OK):

```
{ "success": true, "points_awarded": { "referrer": 350, "referee": 200 } }
```

3. POST /api/v1/loyalty/redeem-calculate

- Purpose: Calculates the maximum points redeemable against a specific course fee.
- Request Payload:

```
{ "student_id": "BIA-2024-9042", "course_fee": 10000, "points_requested": 2000 }
```
- Response (200 OK):

```
{ "max_cap_percent": "30%", "max_discount_aed": 3000, "points_deducted": 2000, "discount_aed": 500, "final_payable_fee": 9500 }
```

Note: If a student tries to deduct more points than their balance or exceeds the tier cap, the API clamps points_deducted to the maximum allowed limit.

5. SHUKRAN-STYLE TIER MANAGEMENT & MULTIPLIERS

To encourage sustained engagement, we implement a Shukran-style tiered progression. Progression is calculated based on the number of verified successful referrals over a rolling 12-month period. Once a tier is achieved, the student unlocks enhanced earning capabilities:

Ø>ÝI Bronze Tier (Classic Membership)

- Qualification: Default level on signup.
- Point Multiplier: 1.0x Rate | Redemption Cap: 30% of course fee.

Ø>ÝH Silver Tier (Engagement Level)

- Qualification: Unlocked at 3 successful referrals.
- Point Multiplier: 1.1x Rate (e.g., 275 pts on base referrals) | Redemption Cap: 35%.

Ø>ÝG Gold Tier (Super Advocate)

- Qualification: Unlocked at 5 successful referrals.
- Point Multiplier: 1.2x Rate (300 pts) | Redemption Cap: 40% | Reward: AED 250 Voucher.

Ø=ÜŽ Platinum Tier (Elite Partner)

- Qualification: Unlocked at 15 successful referrals.
- Point Multiplier: 1.3x Rate (325 pts) | Redemption Cap: 40% | Reward: VIP Alumni Gala Invite.

Calculations and Tier Multipliers Logic:

Standard points are multiplied using the multiplier rate. (For example, if a Silver student completes a SkillShare activity normally worth 100 points, they are credited with 110 points). Point multipliers do not apply to first-referral super bonuses to maintain financial balance.

6. POINTS EXPIRATION ALGORITHMS & MATHEMATICS

The BIA Referral & Loyalty terms state: "Loyalty points are valid for 4 years. After 4 years, 50% of unused points expire, and the remaining 50% stay valid."

To implement this without breaking calculations, we use a First-In-First-Out (FIFO) allocation ledger. Points are deducted from the oldest active transaction record first. When points reach their 4-year limit, the remaining unredeemed balance of that transaction is calculated, and a 50% deduction is applied.

The Nightly Expiration Cron Algorithm (Pseudocode):

```
// Executed daily at 00:05 UTC
SELECT DISTINCT user_id FROM points_ledger WHERE expires_at <= NOW() AND points_remaining > 0
FOR EACH user_id IN users:
    DECLARE expiring_records = SELECT * FROM points_ledger
                                WHERE user_id = user.id AND expires_at <= NOW() AND points_remaining > 0
    FOR EACH record IN expiring_records:
        DECLARE points_to_decay = FLOOR(record.points_remaining * 0.5)
        // Credit the decay transaction
        INSERT INTO points_ledger (user_id, points_change, event_type, description, expires_at,
points_remaining)
        VALUES (user.id, -points_to_decay, "Expiry", "50% Decay of expired credits", NULL, 0)
        // Adjust remaining points on the record
        UPDATE points_ledger
        SET points_remaining = record.points_remaining - points_to_decay,
            expires_at = DATE_ADD(record.expires_at, INTERVAL 2 YEAR) // Extend the remainder
        WHERE ledger_id = record.ledger_id
        // Update user profile cache balance
        UPDATE users SET points_balance = points_balance - points_to_decay WHERE user_id = user.id
```

Why this math is robust:

1. Audit trail is preserved: A negative ledger entry clearly explains points changes.
2. Extensions are automated: The remaining 50% is pushed out by 2 years to prevent looping decays.
3. Performance optimized: Indexes allow searching only the decaying records rather than the whole database.

7. USER EXPERIENCE (UX), TECH STACK & ANIMATIONS (GSAP)

To achieve an app-like premium feel, the student portal is designed using glassmorphic minimalism and powered by the GreenSock Animation Platform (GSAP). Below is a simple explanation of the design language for non-technical stakeholders:

A. Why GSAP? (Sleek Visual Performance)

Unlike heavy video scripts or complex CSS animations that can slow down mobile phones, GSAP is an ultra-fast, professional-grade animation library. It ensures all interface transitions look butter-smooth across all smartphones, tablets, and computers, boosting student satisfaction.

B. Premium Micro-Animations Catalogue

- **Dynamic Points Rollup:** When logging in, points count up dynamically from 0 (like a scoreboard rollup). This creates a sense of reward and accomplishment.
- **Physics-based Progress Bar:** The tier progress bar slides and bounces elastically on load.
- **Interactive 3D Tilt:** The loyalty wallet card tilts and catches light reflections based on the user's mouse cursor movements, providing a tactile, premium feeling.
- **Glowing Milestone Alerts:** When a reward unlocks, it pulses with a subtle emerald glow.

C. Simplified Technology Stack

- **Backend Core:** Laravel (PHP). Highly secure, easily updates points in the MySQL database, and shares hosting directly with BIA's main servers (saving infrastructure costs).
 - **Frontend Interface:** Modern HTML/CSS + GSAP JavaScript. Students can check their wallets directly on their mobile browsers without having to install a separate mobile app.
-

8. WHATSAPP BUSINESS API & LMS INTEGRATIONS

To provide a premium Shukran-like experience, students must be able to interact with the engine from the platforms they use daily: WhatsApp and the BIA LMS (SkillShare portal).

1. WhatsApp Bot Microservice Interface

The BIA Whatsapp bot microservice communicates with the Loyalty Engine API. It decodes incoming webhook commands from the Twilio/360dialog sandbox:

- Command: "/balance" or "check points"

Bot Reply: "Hello Sarah! Your points balance is 1,250 points, equivalent to AED 312.50. You are in Gold Tier (Progress: 5/15 referrals to Platinum)."

- Command: "/referral"

Bot Reply: "Your unique referral link is: <https://loyalty.bradfordia.com/ref/SARAH-9042>. Share it with friends to earn up to 350 points per enrollment!"

2. LMS & SkillShare Activity Sync

When a student completes an academic milestone or course on BIA's internal LMS (e.g. SkillShare), the LMS triggers a POST hook:

POST <https://loyalty.bradfordia.com/api/v1/loyalty/lms-complete>

Headers: { "X-BIA-LMS-Token": "secret_hash" }

{ "student_email": "sarah@email.com", "activity_id": "cma_module_1", "points_value": 150 }

The engine verifies the token signature, calculates the student's tier multiplier, logs the points credit (e.g., +180 points for Gold tier), and sends a WhatsApp confirmation alert.

Note: Rate limiters on the LMS webhook verify that a student cannot earn points for the same course completion activity multiple times.

9. SECURITY PROTOCOLS, ANTI-FRAUD MIDDLEWARE, & COMPLIANCE

A common vulnerability in loyalty programs is point manipulation and self-referrals (referring fake email accounts to generate points). We implement several security checks to protect BIA from financial exploitation:

A. Self-Referral Prevention Middleware

When a referred lead registers, the system records and cross-checks:

- Device Fingerprints: Block registration if both referrer and referee shares the same device token.
- IP Address Clustering: Flags accounts registering multiple referees from the same IP within a 1-hour window.
- Email Similarity Check: Rejects signups where the referee email mimics the referrer (e.g., sarah+test1@email.com).

B. Velocity Control Caps

To prevent script attacks from spamming links across online forums, we enforce velocity limiters:

- Hourly Limit: Max 3 referral submissions per user per hour.
- Monthly Cap: Max 10 successful point accruals from referrals per month. Subsequent referrals go to an Admin Hold Queue for manual review.

C. Ledger Balance Verification Checks

Before any point redemption is approved, the system runs an audit query:

```
SELECT SUM(points_change) FROM points_ledger WHERE user_id = ?
```

The sum of the ledger records must exactly equal the user's current cached points balance. If there is a mismatch, the account is locked instantly, and an alert is sent to BIA admins.

D. UAE Legal Compliance

Under UAE consumer data laws, loyalty points are non-transferable, cannot be exchanged for physical cash, and expire within defined parameters. The terms and conditions are explicitly displayed to students in the portal signup forms.

10. IMPLEMENTATION TIMELINES & QA TEST SUITE

Employing an intern paired with Antigravity AI allows BIA to launch this project in 12 days. Below is the operational schedule and the test suites the intern must execute to verify success:

Intern Deployment Action Plan (12 Days)

- Day 1-2 (Database Setup): Intern runs Laravel DB migrations and registers indexes.
- Day 3-4 (Core API Controllers): Intern builds controllers for points earn, spend, and balances.
- Day 5-6 (Expiration Scheduler): Intern writes daily scheduler console commands.
- Day 7-8 (Webhook Listeners): Intern integrates LMS endpoints and test hooks.
- Day 9-10 (Responsive Portal View): Intern designs responsive CSS mobile dashboards.
- Day 11-12 (UAT & Go-Live): Intern executes tests and deploys production server.

Quality Assurance Testing Checklist

The intern must run these testing suites before project handoff:

1. Unit Test: `point_accrual_test.php`

- Input: Submit referral for MBA program from a Gold Tier student.
- Expected Output: Verify referrer balance increases by 250 points, and referee gets 200 points. Check ledger.

2. Integration Test: `points_expiration_mock_test.php`

- Input: Mock time to 4 years forward. Run console schedule command.
- Expected Output: Check that 50% of the unused balance decays, and ledger records an Expiry transaction.

3. Security Test: `self_referral_block_test.php`

- Input: Trigger lead submission sharing the same device token or email alias.
 - Expected Output: System returns 403 Forbidden. Transaction ledger block verified.
-

